

BANDEEP SINGH

SOON IT WILL BE POSSIBLE TO DO
RAPID TESTS TO FIND A PATIENT'S
GENOTYPE AND GUIDE TREATMENT
WITH THE MOST EFFECTIVE DRUGS.

GENE ANALYSIS AT A DELHI LABORATORY

European populations were not seen among Indians. They were also able to identify a novel gene that is associated with asthma among Indian patients. These would help them identify high-risk individuals and also advise them on preventive steps. Carrying the process forward, the team studied the absorptive ability of populations to the drug salbutamol used to treat asthma. In some people it was found to be not as effective as in others. Understanding the process would help in administering the right amount of doses of the drug and getting away from "the one-size-fits-all" approach.

Somewhat similar work is being done on Diabetes Type II by his colleague Dwaipayan Bharadwaj who is looking for genetic markers that could help doctors identify those susceptible to it long before they come in for treatment. Again here, after examining over 7,000 patients suffering from this type of diabetes, Bharadwaj's study shows that while Indians did have some main gene markers that Europeans had, large amount of them were absent too. The hope is that if they could home in on genetic markers for Diabetes Type II they could warn those Indians at risk early to enable them to take preventive measures and improve their quality of life. Such is the promise of pharmacogenomics that Brahmachari says: "We should soon be able to do rapid tests to determine a patient's genotype and then guide treatment with the most effective drugs apart from reducing adverse reactions."

There is also the growing clamour to resort to gene therapy or the potential of using genes themselves to treat disease or enhance particular traits much like the research into stem cells. While genetics holds up the promise of revolutionising health care, there is need to be cautious about its use. Genetic profiling would reveal some of the innermost secrets that could, for instance, be used by employers to weed out candidates found to be susceptible to certain ailments or diseases. "We deliberately didn't release details of the communities in our studies because of the social ramifications involved," says Mitali Mukerji, IGV consortium co-ordinator. So before you go out and get your genes sequenced—it costs around Rs 2.5 lakh abroad—know the benefits and the cost. ■

“The best way to understand the Indian

gene pool is to compare it to a large banyan tree which has a trunk with branches than twigs and leaves. ”

**Ramaswamy
PITCHAPPAN**

Professor Emeritus, Madurai Kamaraj University

